Semantic segmentation of ultra-high resolution remote sensing images based on fully convolutional neural networks

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***Abstract*—Semantic segmentation has been widely used in various fields of remote sensing; however, ultra-high resolution remote sensing images, due to their extremely high resolution, exhibit more spatial details and cannot be directly segmented using manual segmentation or traditional semantic segmentation methods. We applied fully convolutional deep learning-based neural networks to the semantic segmentation of ultra-high-resolution remote sensing images and at the same time designed an efficient image segmentation strategy to reduce the difficulty of training while ensuring the integrity of the dataset. We conducted experiments on the ultra-high resolution remote sensing dataset Potsdam and demonstrated that the semantic segmentation method based on fully convolutional neural networks can achieve 93.78% MIoU and 97.39% PA on the training set.**

Keywords-Ultra-high resolution; remote sensing; semantic segmentation; fully convolutional neural networks

I. INTRODUCTION

Remote sensing technology has been widely used in agriculture, environmental protection, and other fields. With the development of spatial imaging technology, the spatial resolution of remote sensing images has gradually increased. According to the different spatial resolutions, remote sensing images can be divided into four categories. Low and medium resolution (LTMR) refers to remote sensing data with a spatial resolution greater than 4m [1]. High Resolution (HR) refers to remote sensing data with 1 to 4m spatial resolution, such as China's Gaofen-1 and Gaofen-2 satellites. Very High Resolution (VHR) refers to remote sensing data with a spatial resolution of 0.3 to 1m and is available from most commercial satellites, such as WorldView and QuickBird. Thanks to developments in aerial photogrammetry, it is now possible to obtain ultra-high resolution (UHR) images with a spatial resolution of less than 0.3m from airborne or unmanned aerial platforms [2]. With the increase of spatial resolution, remote sensing images face problems such as complex environments, large scene range, many categories of ground objects in the scene, large intra-class variance and small inter-class variance of ground objects, and fewer imaging spectral bands, which make manual interpretation and traditional image element-based classification methods unable to directly classify high-resolution remote sensing images and make feature identification more difficult. The classification of ultra-high resolution remote sensing images is even more difficult. With the continued success of deep learning in the field of computer vision, image classification models based on convolutional neural networks have emerged. Through the design of various encoders and decoders, both local features and global features can be extracted and restored, and "end-to-end" automatic feature learning can be carried out to achieve automatic decoding of remote sensing images for accurate target identification and classification.

There are many semantic segmentation models based on convolutional neural networks. U-Net [3] is a U-shaped symmetric structure first applied to medical image segmentation, which mainly consists of a contracting path and a symmetric expansive path, and is also an encoder-decoder structure. The contracting path obtains the contextual information of the image through convolution, ReLU, and max pooling, while the layers in the expansive path and the corresponding layers in the contracting path are cropped and fused by “concat”, and then upsampled by deconvolution to precisely locate the segmentation boundary. U-Net uses mirroring of the input images to fill in the missing content, cuts the images into patches for training, and the amount of training data is much larger than the number of training images, which can ensure that the network can maintain invariance and robustness even with fewer samples. In addition, U-Net proposes a weighted loss function to solve the category imbalance problem. FusionNet [4] is a deep neural network based on residual blocks. The whole network looks like a top-down structure instead of a left-right structure of U-Net, using many “burger” structures in between, each “burger” structure consists of one convolutional layer, one residual block, and another convolutional layer. FusionNet uses “sum” instead of “concat” for skip connections. DeconvNet [5] is very similar to SegNet [6] in that both are typical encoder-decoder structures.

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In the encoder, both use the first 13 convolutional layers and 5 max-pooling layers of the VGG16 network to complete the downsampling and in the decoder, both use the upsampling by unpooling. The difference is that DeconvNet uses deconvolution in its decoder, whereas SegNet uses convolution in its decoder, and uses the pooling index in the decoder when unpooling. In addition, DeconvNet uses two fully-connected layers of VGG16 in between the encoder and decoder, which increases the number of parameters in the network and makes training time longer.

Currently, there are relatively few studies related to semantic segmentation of ultra-high resolution remote sensing images. Tian et al.[1] constructed a large-scale ultra-high resolution urban change detection dataset based on deep learning for binary change detection and semantic change detection, named Hi-UCD dataset, which has a total of 40800 pairs of 512×512 patches, 9 land cover categories and 48 semantic changes, and three metrics: binary consistency, change area consistency and no change area consistency, were developed to evaluate the semantic consistency. Sun et al. [7] analyzed the effects of different patch sizes and input resolutions on the segmentation accuracy of ultra-high resolution remote sensing images, proposed a multi-scale collective fusion (MSCF) method to utilize the information from multiple resolutions for end-to-end training, and achieved efficient training on the DeepGlobe dataset. Ramdani et al. [8] described small-scale geometrical properties (i.e. humans) based on ultra-high resolution optical and LiDAR data and illustrated how innovations in remote sensing can improve the ability to draw contours of regions in ultra-high spatial and partial detail for the identification of moving living creatures. Lin et al. [9] used mobile laser scanning (MLS) data as a 3D reference to enrich and calibrate image retrieval while extending MLS coverage with ultra-high resolution imagery, both of which were used in combination for fine-scale 3D biotope mapping.

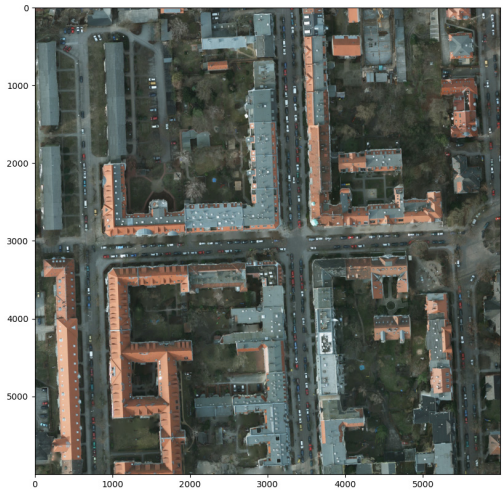
When using deep networks to train ultra-high resolution remote sensing images, implementation is difficult due to hardware constraints such as GPUs. In this thesis, we design an effective image segmentation strategy for cutting ultra-high resolution remote sensing images to reduce the training difficulty while ensuring the integrity of the dataset, and use a deep learning-based fully convolutional neural network to perform end-to-end semantic segmentation of ultra-high resolution remote sensing images, clearly and explicitly describing and demonstrating the feature extraction and feature reduction process of the fully convolutional neural network.

II. Data and Method

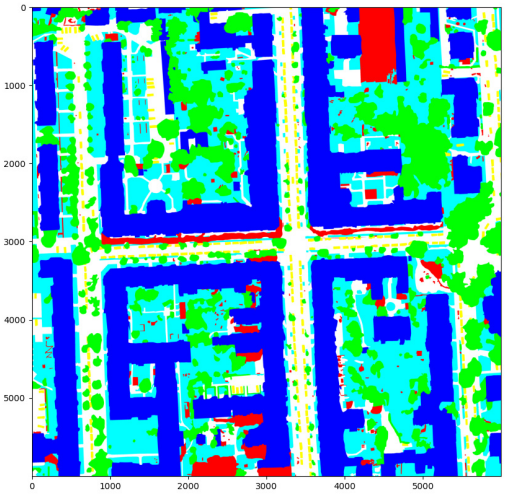
A. Dataset

ISPRS provides two state-of-the-art high-resolution remote sensing image datasets for the Urban Classification and 3D Building Reconstruction Test Project, Vaihingen and Potsdam, both of which cover urban scenes. Vaihingen, a relatively small village, has reached a spatial resolution of 0.09m, which is already a very high-resolution remote sensing image dataset, while Potsdam can resolve 0.05m, which is a typical ultra-high resolution remote sensing image dataset.

We use the Potsdam dataset, which is a typical historical city with large building blocks, narrow streets, and dense settlement structures. 38 images of 6000×6000 pixel UAV images, as shown in Fig. 1, are included in the Potsdam dataset, with three different channels of image data and labeled data, with and without borders. We used the RGB three-channel data with borders. This dataset, like the Vaihingen dataset, was manually classified into the six most common land cover categories, *Impervious surfaces, Building, Low vegetation, Tree, Car, and Clutter/background*.



(a)



(b)

Fig. 1 Potsdam Dataset. (a) True Orthophoto (b)Ground Truth.

As the spatial resolution of remote sensing images increases, spatial detail information becomes richer, allowing us to distinguish features between different objects and obtain precise boundaries of change [1]. At the same time, deep neural networks for ultra-high resolution remote sensing images may encounter greater difficulties in training due to hardware constraints such as GPUs. We trained the deep neural network by cutting the 37 (except top_potsdam_6_7) ultra-high

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resolution remote sensing images in the Potsdam dataset into multiple patches of 512 × 512, as shown in Fig. 2. When cutting, both to maintain the full boundary information and to keep the number of data sets, we set overlap of approximately 10% for each image, and at the end of the cut, when it is detected that each row or column has less than 512 remaining dimensions, a horizontal forward or vertical upward shift is made to ensure that the end of the image is also cut out. We cut out a total of 6253 images, divided into a training set, a validation set, and a test set, containing 4225, 1014, and 1014 images respectively, in a ratio of approximately 60%, 20%, and 20%.

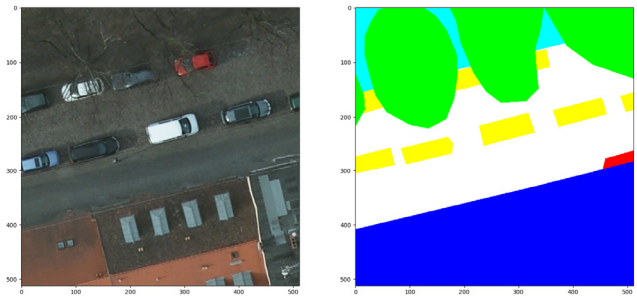


Fig. 2 Image after cutting (512×512)

B. Method

Fully convolutional networks (FCN) [10], are the most fundamental and classical deep learning-based approach to semantic segmentation of images. Many successful semantic segmentation networks that have emerged subsequently have been improved based on fully convolutional networks. Fully convolutional neural networks replace the last three fully connected layers in popular deep network models for image classification such as VGG16, AlexNet, and GoogLeNet with multi-channel convolutional layers of equal vector length corresponding to 1 × 1 convolutional kernels, achieving fully convolution of the network and making the final output a picture of equal size to the input image rather than a one-dimensional vector. Fully convolutional neural networks can label each pixel on the input image for classification, a pixel-level recognition, and achieves end-to-end semantic segmentation by taking into account semantic and positional information, i.e. global and local information, through a unique skip structure. Fully convolutional neural networks consist of two parts: feature extraction by downsampling and feature reduction by upsampling and feature fusion. The FCN retains local features in the downsampling process, and in the upsampling process, the finally generated global features are fused with the local features of the corresponding layers in the downsampling through a skip structure before multiple upsampling, resulting in three semantic segmentation deep networks, FCN-32s, FCN-16s, and FCN-8s.

1) Feature Extraction

The feature extraction process of FCN-32s, FCN-16s, and FCN-8s is the same, all of them extract global feature information through continuous convolution and max pooling downsampling, where the convolution kernel for convolution is 3 and the pooling kernel for convolution is 2. Using VGG16 as the backbone and a 512×512×3 Potsdam remote sensing image as an example, the feature extraction process of the fully convolutional neural network has a total of 5 layers of convolution and 5 layers of pooling, the specific process is shown in Fig. 3.

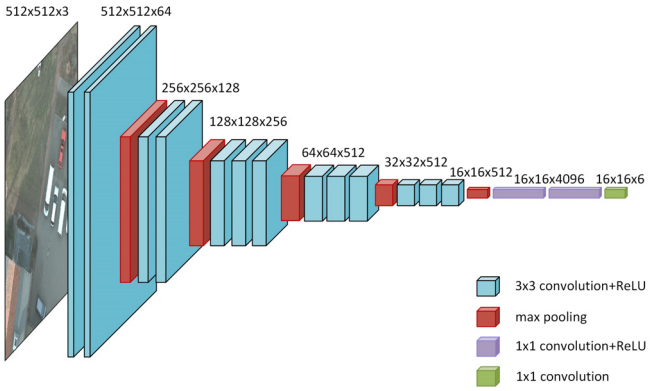


Fig. 3. Feature extraction process for fully convolutional neural networks

In the first layer, the input image is convolved through two layers to obtain a 64-channel feature map, i.e. the size of the feature map becomes 512×512×64.

In the second layer, the feature map obtained from the first layer is first subjected to max pooling to complete the first downsampling, with the width and height of the feature map becoming 1/2 of the original, and then two convolutions are performed to transform the number of channels to 128, with the final feature map size becoming 256×256×128.

In the third layer, the feature map obtained from the second layer is first subjected to max pooling to complete the second downsampling, with the width and height of the feature map becoming 1/2 of the original, i.e. 1/4 of the input image, and then three convolutions are performed to transform the number of channels to 256, with the final feature map size becoming 128×128×256.

In the fourth layer, the feature map obtained from the third layer is first subjected to max pooling, completing the third downsampling, with the width and height of the feature map becoming 1/2 of the original, i.e. 1/8 of the input image, and then three convolutions are performed to transform the number of channels to 512, with the final feature map size becoming 64×64×512.

In the fifth layer, the feature map obtained from the fourth layer is first subjected to max pooling, and the fourth downsampling is completed, with the width and height of the feature map becoming 1/2 of the original, i.e. 1/16 of the input image, and then three convolutions are performed, with the number of channels remaining unchanged at 512, resulting in a final feature map size of 32×32×512.

Finally, the feature map in the fifth layer is max-pooled again to complete the fifth downsampling, and the width and height of the feature map become 1/2 of the original, i.e. 1/32 of the input image, with the same number of channels, still 512, and the final feature map size becomes 16×16×512. So far, the fully convolutional neural network completes the local feature

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extraction, after which the number of channels in the feature map is changed to the number of categories in the dataset, which is 6 in the Potsdam dataset, by convolving the three convolutional kernels to 1.

The size of the feature map at this point is 1/32 of the input image. When the fully convolutional neural network performs semantic segmentation, the size of the output image is the same as the input image, which requires feature reduction using deconvolution in between.

2) Feature Reduction

Semantic segmentation suffers from the problem that global information and local information have internal contradictions, this is because, in the process of downsampling, the network gets deeper and deeper, local information will be gradually lost, and both are decomposed into global information, the two can be seen as the relationship between this and the other. In the fully convolutional neural network, a unique skip connection structure is designed to retain the previous local features during downsampling and then fuse them with the global features generated by the last convolutional layer for upsampling reduction, so that the feature map is restored to the same size as the input image, thus ensuring that the input and output images are equal in size. The feature reduction process for FCN-32s, FCN-16s, and FCN-8s is shown in Fig. 4.

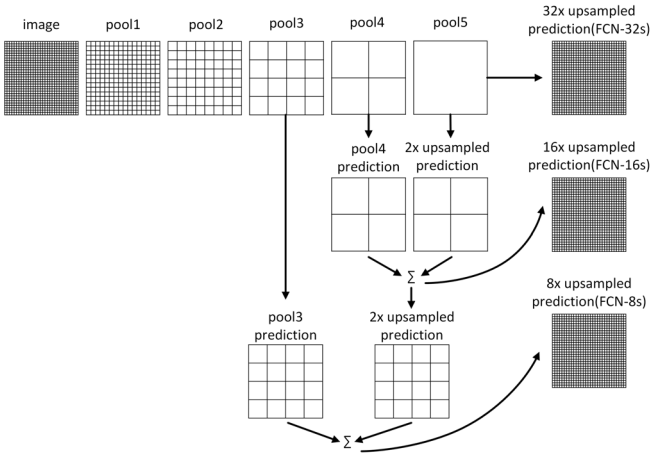


Fig. 4. Feature reduction process for fully convolutional neural networks

FCN-32s is a direct 32x deconvolution upsampling of the feature map (of the same size as after the fifth maximum pooling, i.e. 1/32 of the input image) from the last (last in Fig. 3) convolution output of the feature extraction process to obtain a feature map of the same size as the input image; this approach does not use skip connection.

FCN-16s is the feature map output from the last convolution in the feature extraction process is first upsampled by 2x deconvolution into a feature map of 1/16 the size of the input image, then fused with the feature map after the fourth max pooling, i.e. skip connection, and finally upsampled by 16x by deconvolution to obtain a feature map of the same size as the input image.

FCN-8s is to fuse the image after feature fusion in the FCN-16s network (size is 1/16 of the input image) into a feature map of 1/8 size of the input image by 2x deconvolution upsampling first, then fused with the feature map after the third maxi pooling to achieve the second skip connection, and finally 8x upsampling by deconvolution to get a feature map of the same size as the input image.

III. Results & Discussion

PyTorch is now a popular framework for training deep learning models, encapsulating a large number of features with powerful GPU-accelerated tensor computing power. In our experiment, the programming language is Python 3.7, the deep learning framework is PyTorch 1.13 GPU version, torchvision 0.14.1, torchaudio 0.13.1, the computer configuration is Intel(R) Xeon(R) E5-2650 v4@2.20GHz(X2) CPU, NVIDIA Tesla V100-PCI-32GB graphics card, 112GB DDR4 2666MHz RAM. We set up 300 training epochs with an image input size of 512×512, a dataset category size of 6, and a batch size of 8. The optimizer used Adam with an initial learning rate of 1×10^{-4} and adjusted the learning rate to half of the original rate after every 50 epochs.

Common evaluation metrics for semantic segmentation include Pixel Accuracy (PA), Mean Pixel Accuracy (MPA), Mean Intersection over Union (MIoU), and Frequency Weighted Intersection over Union (FWIoU). We applied FCN-32s, FCN-16s, FCN-8s, and U-Net on the ultra-high resolution remote sensing image dataset Potsdam for comparative experiments on semantic segmentation respectively. After completing 300 epochs of training, we obtained a comparison of the performance metrics of the four networks on the training set, validation set, and test set as shown in TABLE I, TABLE II, and TABLE III. In addition, we also compared the IoU of the four networks on the six categories of the training set, validation set, and test set as shown in TABLE IV, TABLE V, and TABLE VI.

On the training set, the PA, MPA, MIoU, and FWIoU of the three fully convolutional neural networks reached 97%, 96%, 93%, and 94% or more respectively, all higher than U-Net, and all performed well in segmentation performance metrics. On the validation set, the three networks achieved 87%, 84%, 74%, and 77% or more for each performance metric respectively. On the test set, the three networks achieved 78%, 76%, 59%, and over 64% for each performance metric respectively. As can be seen in TABLE II and TABLE III, the FCN-8s achieved the highest values in all metrics on both the validation and test sets. In addition, all three fully convolutional neural networks achieved good segmentation accuracy in the comparison of IoU for each category in the dataset. On the training set, the three networks achieved the highest IoU values of 95.86%, 97.77%, 93.21%, 93.79%, 95.89%, and 86.19% for the six categories respectively, and FCN-8s achieved the highest values for the validation and test sets for four categories with the highest values in three categories, as shown in Table IV. On the validation set, the lowest IoU values were even higher than those of U-Net by 2.23%, 2.8%, 1.47%, 2.89%, and 4.96% for all three fully convolutional neural networks for six categories except *Car*, as shown in Table V.

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TABLE I. COMPARISON OF PERFORMANCE METRICS OF FOUR NETWORKS ON THE TRAINING SET

Network	PA(%)	MPA(%)	MIoU(%)	FWIoU(%)
FCN-32s	97.38	96.49	93.76	94.92
FCN-16s	97.39	96.47	93.78	94.95
FCN-8s	97.33	96.41	93.58	94.83
U-Net	97.00	96.04	92.92	94.21

TABLE II. COMPARISON OF PERFORMANCE METRICS OF FOUR NETWORKS ON THE VALIDATION SET

Network	PA(%)	MPA(%)	MIoU(%)	FWIoU(%)
FCN-32s	87.00	84.42	74.52	77.51
FCN-16s	87.15	84.57	74.74	77.81
FCN-8s	87.18	84.75	74.83	77.82
U-Net	85.33	82.73	71.97	74.96

TABLE III. COMPARISON OF PERFORMANCE METRICS OF FOUR NETWORKS ON THE TEST SET

Network	PA(%)	MPA(%)	MIoU(%)	FWIoU(%)
FCN-32s	78.33	76.35	59.09	64.64
FCN-16s	78.14	76.67	59.27	64.36
FCN-8s	78.48	77.11	59.63	64.85
U-Net	77.21	75.32	57.72	63.38

TABLE IV. COMPARISON OF THE IOU OF EACH CATEGORY ON THE TRAINING SET FOR FOUR NETWORKS

Network	IoU(%)			
	FCN-32s	***FCN-16s***	***FCN-8s***	***U-Net***
Impervious surfaces	95.79	95.86	95.71	95.15
Building	97.72	97.77	97.69	97.10
Low vegetation	93.21	93.21	93.16	92.60
Tree	93.94	93.92	93.79	93.10
Car	95.74	95.89	95.33	95.08
Clutter/Background	86.19	86.04	85.83	84.48

TABLE V. COMPARISON OF THE IOU OF EACH CATEGORY ON THE VALIDATION SET FOR FOUR NETWORKS

Network	IoU(%)			
	FCN-32s	***FCN-16s***	***FCN-8s***	***U-Net***
Impervious surfaces	80.94	81.15	81.32	78.71
Building	89.52	89.63	89.63	86.72
Low vegetation	68.40	68.22	69.19	66.75
Tree	71.61	71.70	71.92	68.72
Car	81.95	83.04	83.21	82.18
Clutter/Background	54.71	54.69	53.71	48.75

TABLE VI. COMPARISON OF THE IOU OF EACH CATEGORY ON THE TEST SET FOR FOUR NETWORKS

Network	IoU(%)			
	FCN-32s	***FCN-16s***	***FCN-8s***	***U-Net***
Impervious surfaces	65.06	64.59	65.04	63.87
Building	74.62	74.11	74.67	73.81
Low vegetation	54.82	54.72	55.29	53.43
Tree	65.12	65.25	65.78	63.48
Car	65.33	66.63	66.23	66.42
Clutter/Background	29.61	30.32	30.77	25.33

In Fig. 5-9, we show how the loss function values, PA, MPA, MIoU, and FWIoU change during training for the four networks. The comparison shows that the values of the loss functions of the three fully convolutional neural networks converge significantly more blocky than those of U-Net, and are also significantly faster than U-Net in terms of the rapid improvement in the three metrics of PA, MIoU, and FWIoU. Additionally, the loss values and various performance metrics of the four networks all leveled off after 50 epochs and the loss values converged, but after 100 and 150 epochs there was still a small change from a small decrease in loss values and a small increase in the remaining metrics, and after 300 epochs the change was very subtle and basically no more change. In summary, semantic segmentation of ultra-high resolution remote sensing images using fully convolutional neural networks performs well in all of the performance metrics.

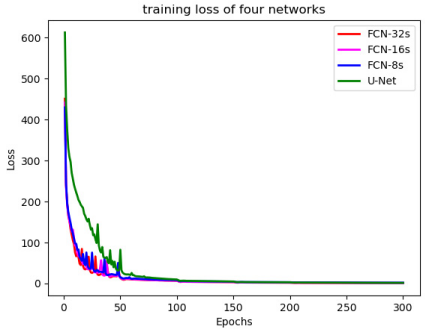


Fig. 5 Variation of loss value during training of four networks

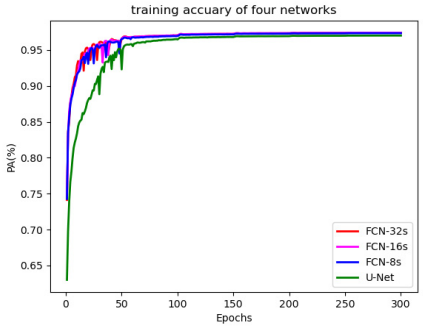


Fig. 6 Variation of PA during training of four networks

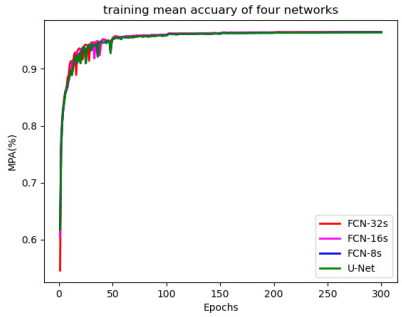


Fig. 7 Variation of MPA during training of four networks

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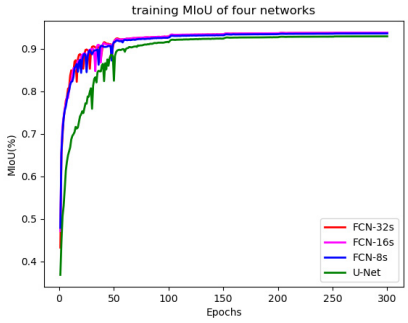


Fig. 8 Variation of MIoU during training of four networks

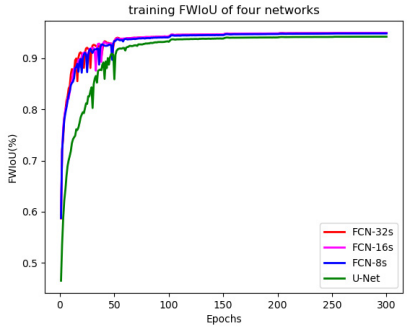


Fig. 9 Variation of FWIoU during training of four networks

We have selected five representative remote sensing images from the segmented Potsdam dataset and show their segmentation maps using each of the four network outputs, as shown in Fig. 10.

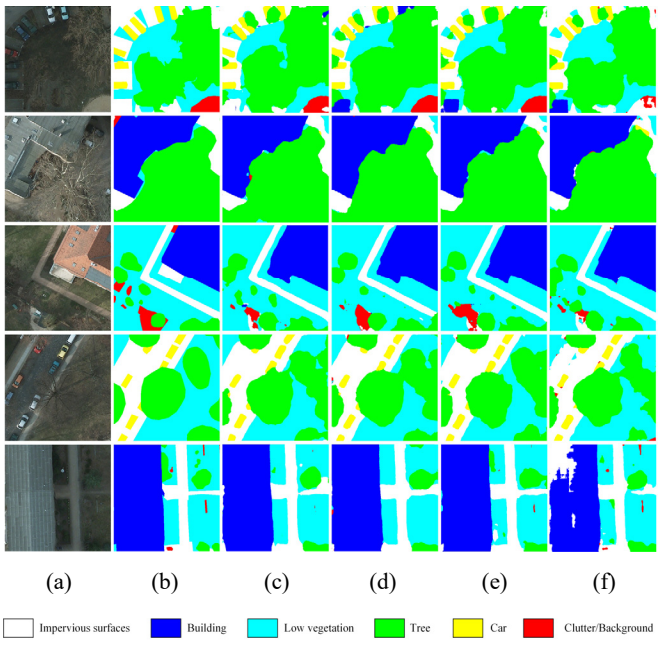


Fig. 10 Semantic segmentation results in the Potsdam dataset with FCN-32s, FCN-16s, FCN-8s, and U-Net, respectively. Different colors represent different labels. (a)Image (b)Ground Truth (c) FCN-32s (d) FCN-16s (e) FCN-8s (f) U-Net.

IV. Conclusions

In this paper, we have effectively segmented the ultra-high resolution remote sensing images with a resolution of 6000×6000 from the Potsdam dataset, which not only allows the full use of all images and guarantees the segmentation boundary information during training, but also clearly describes and demonstrates the feature extraction and feature reduction process of the fully convolutional neural network, and finally applies it to the semantic segmentation of ultra-high resolution remote sensing images, completing enough training experiments for many epochs. The experimental results show that the fully convolutional neural network can achieve semantic segmentation of ultra-high resolution remote sensing images with high quality and good results in all performance metrics. In future work, we will try to incorporate spatial pyramids and dilated convolution into the model to further improve the accuracy and generalization capability of the model.

Acknowledgments

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